

Financial education and responsible consumption in undergraduate management students

Dolores Gallardo-Vázquez^{*}, José Luis Miralles-Quirós, María Mar Miralles-Quirós

Departamento de Economía Financiera y Contabilidad, Facultad de Ciencias Económicas y Empresariales, Universidad de Extremadura, Av. Elvas, s/n, 06006, Badajoz, Spain

ARTICLE INFO

JEL classification:

G40
G41
G53
G590
M14
M21

Keywords:

Financial education
Responsible consumption
Undergraduate students
Lifelong learning

ABSTRACT

In recent years, many countries have implemented policies to promote Financial Education (FE) among citizens as a mechanism for boosting economic development and social prosperity. Since 2008, Spain has had a national FE strategy that primarily promotes basic FE among students in compulsory education. This study analyzes first-year university students' perception of the FE acquired in earlier educational stages and its practical application in their personal lives through the acquisition of responsible consumption habits. A survey is carried out in which students are asked how certain basic financial attributes (savings, investment, budgeting, expenses, income and quality of life), translate into everyday life, and their relationship is analyzed with their perception of the importance of FE, and certain values such as honesty, responsibility and solidarity, by applying a PLS-SEM methodology. The results allow us to extract proposals for improving public policies on FE in university studies and in earlier stages.

1. Introduction

Financial Education (FE) has become particularly important in today's society because adult citizens face countless financial decisions in their day-to-day life and because they seem to be particularly ill-prepared for this. As Santos (2017) and Moreira, De Sá Teixeira, Cordeiro, and Santos (2021) point out, due to the systematic transformations in contemporary societies, all individuals are increasingly confronted with making financial decisions, whether to finance basic consumer goods or to meet fundamental needs linked to education, housing or retirement (Gallego-Losada, Montero-Navarro, Rodríguez-Sánchez, & González-Torres, 2022). However, successive international studies have revealed that significant population segments in the developed world do not have the necessary knowledge or understanding to cope with elementary financial decisions (OECD, 2005; Lusardi & Mitchell, 2007; Lusardi & Mitchell, 2011a,b; Standard & Poors, 2015; Lusardi & López, 2016; OECD/INFE, 2016; Moreno-Herrero, Salas-Velasco, & Sánchez-Campillo, 2018a; Moreno-Herrero, Salas-Velasco, & Sánchez-Campillo, 2018b; Lusardi, Michaud, & Mitchell, 2020, among others).

Specifically, the first international study conducted by the Organisation for Economic Co-operation and Development (OECD) in 2005 already warned of the negative consequences of not planning personal finances sufficiently in advance or having inadequate consumption habits. It therefore urged countries to implement national FE policies and strategies to avoid future problems (OECD, 2005). Although the international financial crisis that began in the following years was not caused by the world population's lower

^{*} Corresponding author.

E-mail addresses: dgallard@unex.es (D. Gallardo-Vázquez), miralles@unex.es (J.L. Miralles-Quirós), marmiralles@unex.es (M.M. Miralles-Quirós).

level of FE, the consensus is that this deficient level of financial knowledge exacerbated the negative consequences, and the crisis drove developed economies to begin implementing national FE plans and strategies (Domínguez, 2017).

The growing concern for FE-related issues at the political level has been accompanied by greater interest from the academic point of view. Although there are early works such as those by Bernheim, Garrett, and Maki (2001, 1998) and Braunstein and Welch (2002) on the U.S. population's level of FE at the end of the last century, most international studies emerged after the global financial crisis. These studies are characterized by analyzing the determinants of FE or its impact on individuals' financial behavior as a function of their level of FE.

In the first group of analyses, the work of Lusardi and Mitchell (2007, 2011b, 2014) stands out. These authors were pioneers in analyzing the population's knowledge of FE based on three questions, commonly referred to as the Big Three: interest rates, inflation and risk diversification. Their results document very low levels of FE in the world population, with small differences between countries and population groups, and the most vulnerable segments being the young and elderly, women and citizens with a low level of education and per capita income (Lusardi et al., 2020). On the other hand, the works of Van Rooij, Lusardi, and Alessie (2011), de Bassa Scheresberg (2013), Jappelli and Padula (2013), Atkinson, Messy, Rabinovich, and Yoong (2015) and Moreira Costa, DeSáTeixeira, CordeiroSantos, and Santos (2021), among others, stand out, documenting that consumers with higher levels of financial literacy are more likely to accumulate savings, invest in stock markets or take out private pension plans for retirement. In contrast, consumers with lower levels of financial literacy tend to have detrimental financial behavior with high levels of indebtedness (Lusardi & Tufano, 2015), low savings and inadequate retirement planning (Lusardi & Mitchell, 2011a; Gallego-Losada et al., 2022).

Previous empirical evidence can also be divided according to the population segment analyzed. Although most studies focus on the analysis of the adult population (between the ages of 18 and 65), those that examine the young population that receives financial training in compulsory education also stand out. In this area, the work carried out by Lusardi and López (2016) on the United States, and Moreno-Herrero et al. (2018a and 2018b) on an international level, stand out, highlighting the low level of financial literacy of the students analyzed despite the implementation of public policies on FE.

From the onset, the OECD (2005) pointed out the need for national FE strategies targeting the young population since it considered that FE is more effective the earlier it is implemented. Following this criterion, Mandell and Schmid (2009) believed that the financial knowledge acquired in school remains latent in the minds of young people until their adulthood, when they will have financial resources and apply the financial knowledge acquired in school. In contrast, other authors consider the possibility that financial concepts learned in the very early ages could be easily forgotten if they are not used in the immediate future (Domínguez, 2017).

On the other hand, Villalba, López-Lezama, and Muñoz-Galeano (2017) and Gallardo-Vázquez, Miralles-Quirós, and Miralles-Quirós (2023) emphasize that FE should be an ongoing process that begins in school with adequate training but does not end until adults master and adequately apply financial concepts in their daily lives and professions. Following these authors, we consider essential to examine the necessary relationship between educational stages. To that end, we focus on the analysis of first-year university students. More precisely, the aim of this research is to analyze their perception of the relevance of FE, formed in previous educational stages, as well as the practical application of certain financial aspects in their personal lives.

Meanwhile, we cannot forget to mention the complexity of the financial environment and the risk involved in making decisions, especially for young people (Razen, Huber, Hueber, Kirchler, & Stefan, 2021), determining the well-being, health and life of individuals (Goyal & Kumar, 2021). Very recently, Molina-García, Diéguez-Soto, Galache-Laza, and Campos-Valenzuela (2023) have examined the relationship between FE and risk propensity in a university environment, based on the multidimensional nature of FE, observing that both financial knowledge, attitudes and behavior affect risk propensity. This way, a gap is observed in the literature regarding the level of FE knowledge possessed by young people, capable of determining their future decisions and actions. Given this, we have delved deeper into it, trying to answer the following research questions (RQ):

- **RQ1:** Do students identify the fundamental attributes that determine their level of knowledge in FE?
- **RQ2:** Do students identify the attributes of responsible consumption?
- **RQ3:** Do students relate the level of FE with the responsible consumption they should do?

In order to answer the above research questions, and considering the necessary relationship between educational stages, this study analyzes the perception that first-year university students have of the FE acquired in previous stages and the practical application in their personal lives. In this sense, we view university students as adult citizens who are receiving academic training to become qualified professionals in the near future and who are beginning to face financial decisions in their daily life. These decisions will not foreseeably include the decision to open a pension plan for retirement, but they will include consumption decisions. Specifically, we analyze whether these students have responsible consumption habits and whether they are linked to their appreciation of the importance of FE and its attributes. Ultimately, students need to receive responsible and financial management education that encompasses the skills necessary to make financial decisions in a global and sustainable context (Wihlenda, Brahm, & Habisch, 2023).

For this purpose, a survey is conducted among first-year students in different majors at the University of Extremadura, a Spanish public university, during the 2021-22 academic year. We should take into account that in Spain the competences of education are transferred to the regions and, although there are common national guidelines, the practical application may differ between regions. In this study we analyze the specific case of the region of Extremadura.

In the survey, students are first asked about their overall perception of FE. Second, questions are asked about basic financial aspects such as saving, budgeting, spending, income, investment, and quality of life. These are the attributes that Ortega, Pino, Merino, and Ledrado (2009) identified for the implementation of a FE plan in compulsory secondary education in Spain as part of the Spanish FE strategy. Specifically, it was decided to incorporate these financial attributes in a cross-cutting manner in several subjects in order to

enhance financial knowledge, skills and abilities that would allow students to understand the importance of key concepts, as well as to be able to extrapolate this knowledge and skills to their daily, personal and family life (Gascó & Estruch, 2022; Ortega et al., 2009). Finally, questions are asked about responsible consumption habits in order to analyze how these concepts are applied in daily practice. The survey responses are analyzed by applying a PLS-SEM model, which allows us to establish the relationship between all of the questions.

This type of research allows us to delve into the relationship between perception, knowledge and responsible financial habits. It also allows us to ascertain whether young university students make responsible financial decisions, if they have developed a savings culture, if they understand the importance of adequate economic planning in order to manage their personal and professional economy in the near future, regardless of their profession. The results obtained can be used to propose improvements for public policies in two ways: lines of action can be established for the following university courses, and improvements can be proposed in the area of secondary education so students reach university with a higher level of FE.

This paper is divided into five sections. The second section contains the theoretical basis and the development of hypotheses, the third section describes the methodology used, the fourth section presents the results, and the fifth section lists the discussion and conclusions.

2. Literature review and hypothesis development

2.1. Attributes of financial education

As a first step, it is necessary to specify what is meant by FE. Specifically, we take the most widely accepted definition, which is that of the OECD: "the process by which financial consumers or investors improve their understanding of financial products, concepts and risks and, through information, education and/or objective advice, develop the skills and confidence necessary to become more aware of financial risks and opportunities, make informed decisions, know where to turn for help and take any effective action to improve their financial well-being" (OECD, 2005, p. 26).

Based on this definition, the three key elements of FE according to the European Commission (2007) are: acquiring financial knowledge, developing financial competencies and having a responsible financial attitude leading to individual financial well-being. These three key elements (knowledge, attitudes and financial behaviors) determine the multidimensional nature of FE (Johan, Rowlingson, & Appleyard, 2021; Molina-García, Cisneros-Ruiz, López-Subieres, & Diéguez-Soto, 2023). This implies that national FE programs should teach young people how to use financial knowledge to make informed, intelligent and responsible decisions. This is done by establishing, from a very early age, habits associated with certain attributes of FE.

As we mention in the introduction section, the initial plan adopted by the Spanish authorities in order to comply with the national FE strategy, focused on young people. Thus, it was decided to incorporate aspects such as savings, investment, budgeting, spending, income and quality of life in a cross-cutting manner in several subjects of secondary education to promote training in values related to money management (honesty, responsibility, compliance with deadlines and solidarity) and to enable the acquisition of the necessary knowledge to later develop in the financial world (Gascó & Estruch, 2022; Ortega et al., 2009).

Meanwhile, there is extensive literature documenting the close relationship between FE and these basic concepts. Specifically, Bernheim, Garrett, and Maki et al. (2001) and Braundstein and Welch (2002) were the first to highlight the importance of encouraging young people to save as a means of preventing financial problems in adulthood. Subsequently, Ortega et al. (2009), Samper, Pinto, Arenas, and Roca (2010), Jappelli (2010), Gale and Levine (2011) and Atkinson and Messy (2013), among others, point out that it is not only necessary to promote a savings culture, but also an investment culture. Specifically, these authors argue that by promoting investment habits, individuals become accustomed to plan accordingly to achieve their future goals. Individuals acquire the habit of analyzing their income and expenses, and therefore of prioritizing necessary expenses over superfluous ones, and always within the range of income earned. Finally, these authors argue that an adequate FE should lead to an adequate investment of savings to ensure the best returns with an adequate level of risk.

More recent studies, such as those by Cull, Ehrbeck, and Holle (2014), Raccanello and Herrera Guzmán (2014), Garg and Singh (2018), state that individuals with a higher level of FE achieve a better quality of life because it allows them to make informed and wise financial decisions. It also contributes to resilience in the face of financial instability and even to escaping poverty. On the contrary, a low or non-existent level of FE can lead to over-indebtedness that ends up affecting the household economy and, with it, decisions about work, parenthood, housing and even retirement, which are associated with people's well-being and quality of life (Lusardi & Tufano, 2015). University students face new responsibilities when they become independent from parents, having to correctly manage their finances and consumption, expenses and income, and prepare their own budget, while facing a more complex and sophisticated financial system (Johan et al., 2021). Finally, EF has positive consequences for the financial system and the economy in general (Molina-García, Cisneros-Ruiz, et al., 2023), despite some challenges such as unemployment and uncertain economic prospects (Sutter, Weyland, Untertrifaller, & Froitzheim, 2020).

On the other hand, international institutions, such as the Center for Financial Services Innovation in a 2016 report on financial health, also support these arguments by pointing out that what consumers do today in terms of spending, saving, borrowing, and planning can impact their ability to take advantage of future opportunities and, therefore, their well-being and quality of life (CFSI, 2016).

Finally, taking into account the Spanish FE strategy launched in 2008, which from the beginning established as a priority to integrate the study of financial aspects such as saving, budgeting, spending, income, investment and quality of life in certain secondary education subjects (Gascó & Estruch, 2022; Ortega et al., 2009), it is expected that the university students surveyed have acquired basic

knowledge and adequate habits regarding these aspects and, consequently, have an adequate perception of the importance of FE. Based on these arguments, the following hypotheses are put forward:

- H1a.** Better knowledge and attitude towards savings are associated with a greater perception of the importance of FE.
- H1b.** Better knowledge and attitude towards budget are associated with a greater perception of the importance of FE.
- H1c.** Better knowledge and attitude towards expenditures are associated with a greater perception of the importance of FE.
- H1d.** Better knowledge and attitude towards income are associated with a greater perception of the importance of FE.
- H1e.** Better knowledge and attitude towards investment are associated with a greater perception of the importance of FE.
- H1f.** Better knowledge and attitude towards quality of life are associated with a greater perception of the importance of FE.

2.2. Attributes of responsible consumption

Learning responsible management from the education received constitutes a multidimensional responsibility that goes beyond the transmission of knowledge itself (Wihlenda et al., 2023), incorporating skills, values and the ability to decide in life. This set of values and attitudes are decisive when deciding on consumer behavior, which must reflect the individual's conscience, their obligations and their ethics (Song & Kim, 2018). Responsible consumption implies, in principle, that individuals start with a conscious vision of their needs, are informed and compare different options before making the decision to purchase a good or service. In terms of financial behavior, some research proposes a comprehensive measure that includes cash management, savings and investments, credit use, insurance, and consumption management (Xiao & Dew, 2011).

However, the term "responsible consumption" could be broadened by associating it with sustainable, ethical, and supportive consumption, as per Ortega et al. (2009). In this regard, it should be noted that sustainable consumption, on the other hand, is that which satisfies current needs without harming future generations. Ethical consumption is that which takes into account social implications and values fairer and more environmentally friendly options. Finally, solidarity consumption is that which takes into account regional economic development or that of the poorest populations (Gallardo-Vázquez et al., 2023).

Based on this definition, we can establish a direct relationship between responsible consumption and the attributes of savings, investment, budget, expenses, income and quality of life. Specifically, responsible consumption does not mean to stop buying or to buy cheaper, but to analyze and seek the most convenient option in terms of cost and benefit. Therefore, responsible consumption contributes to establishing the habit of saving and planning personal finances by reasonably avoiding unnecessary expenses. This will favor investment possibilities and a better quality of life.

Braunstein and Welch (2002) were the first to highlight the need to acquire thoughtful consumption habits to avoid unnecessary and burdensome financial decisions. Later, Lusardi (2014) studied savings and consumption decisions, noting that a rational and well-informed individual will under-consume in times of high income. This implies establishing a savings capacity that will allow consumption in times of lower income, such as in retirement.

From an institutional point of view, the European Economic and Social Committee (EESC) also emphasizes the importance of Europeans having responsible consumption habits for financial products. Specifically, it states that a sounder, safer and more transparent financial system requires the participation of informed, responsible, and committed consumers in developing their financial skills (EESC, 2016).

More recently, Barbic, Lucić, and Chen (2019) have documented that wise consumption decisions determine individuals' well-being and quality of life. Specifically, they state that responsible financial consumption behavior affects not only people's finances but also their social status and emotional state. Conversely, the failure to manage personal finances responsibly can have serious long-term consequences for individuals and society as a whole.

Taking into account Spain's FE strategy that includes in secondary education the study of essential aspects such as saving, investment, budgeting, spending, income and quality of life, it is foreseeable that the university students surveyed have acquired knowledge and attitudes that tend to develop responsible consumption habits (Preuss et al., 2023). Based on these arguments, the following hypotheses are put forward:

- H2a.** A better knowledge and attitude towards savings are associated with a greater ability to consume responsibly.
- H2b.** A better knowledge and attitudes towards budget are associated with a greater ability to consume responsibly.
- H2c.** A better knowledge and attitude towards spending are associated with a greater ability to consume responsibly.
- H2d.** A better knowledge and attitude towards income are associated with a greater ability to consume responsibly.
- H2e.** A better knowledge and attitude towards investment are associated with a greater ability to consume responsibly.
- H2f.** A better knowledge and attitude towards quality of life are associated with a greater ability to consume responsibly.

2.3. Financial education and responsible consumption

Given the above, a direct relationship between the level of FE and responsible consumption habits can be expected. The 2005 OECD report states that without a certain level of financial literacy, consumers may not purchase the financial products and services they

need and/or may purchase unnecessary or inappropriate financial products. Furthermore, it adds that FE can benefit consumers by helping them to be better informed about financial products and services and, therefore, to make appropriate and responsible decisions (OECD, 2005).

However, despite the multidimensional nature that research on FE has acquired, we found no empirical studies that directly analyze the relationship between FE and responsible consumption. We only observed the theoretical approach by Muñoz-Céspedes, Ibar-Alonso, and de Lorenzo Ros (2021), who argue that FE favors sustainable behavior and, therefore, can establish responsible consumption habits.

On the other hand, it is important to consider other types of studies such as those conducted by Allgood and Walstad (2011), LaBorde, Mottner, and Whalley (2013) and Xiao, Chen, and Sun (2015). These authors analyze the actual financial behavior of individuals but also consider the subjective dimension of FE (i.e., people's perception of FE and its relationship with certain financial skills and behaviors). Specifically, these authors note that the perception of financial knowledge is greater than actual financial knowledge. In contrast, the perception of financial knowledge is a determinant of an individual's financial behavior. Taking all of these issues into consideration and applying them to the field of responsible consumption, we propose the following hypothesis:

H3. A greater perception of the importance of FE is associated with a greater ability to consume responsibly.

Fig. 1 presents the conceptual model defined and the hypotheses formulated between the constructs.

3. Materials and methods

3.1. Variables, item generation and design of the perceptions questionnaire

We address the analysis of the perception of certain variables that contribute to an improvement in Financial Education (FE) as well as enabling the practice of responsible consumption (RC). All variables are composite lower order constructs (LOC) Mode A. It is necessary to understand the importance of the following financial concepts: savings (S), budget (B), expenses (E), income (I), investment (IN) and quality of life (Q), which are defined below. At the same time, a specific and very broad scale of indicators, built from the theoretical review of FE has been designed, and that make up the questionnaire under study, initially composed of 52 items divided into 9 blocks, beginning with the data corresponding to the profile of respondents:

- Savings (S): the portion of income that we do not spend and that we save for future contingencies or to invest to try to generate more wealth.
- Budget (B): the quantification of the income and expenses that a person expects to have during a certain time period.
- Expenses (E): consumption of a good or service in exchange for consideration, generally in the form of monetary payment.
- Income (I): amount received from the sale of a good or service, generally in the form of a monetary payment.
- Investment (IN): the use of money in an activity or business with the aim of obtaining future income that allows the generation of greater wealth.
- Quality of life (Q): the set of circumstances that guarantee a person's well-being.
- Financial Education (FE): the combination of financial knowledge, financial skills and a responsible financial attitude leading to individual financial well-being.
- Responsible, healthy, ethical, sustainable and supportive consumption (RC): a type of consumption in which the use of products and services covers basic needs and provides a better quality of life.

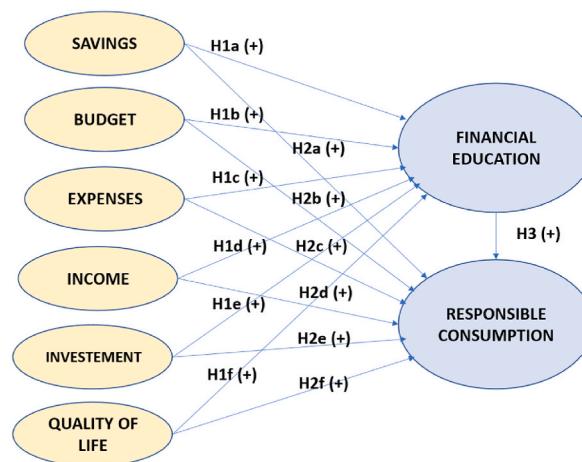


Fig. 1. Conceptual model and hypotheses.

Source: Authors

3.2. Data and characteristics of the sample

The indicators requested the perception of first-year undergraduate students, as they are the closest to entering university education. Prior to the selection of the final data and in order to know if the defined questionnaire was understood by the students, a random sample of 25 students was selected with whom a pretest was carried out. These students calmly read the entire questionnaire, making some observations and confirming that it was perfectly understandable. Subsequently, the clarification of these observations was carried out, obtaining the definitive questionnaire, which was the one that was passed in all the first-year undergraduate classrooms.

For the final data collection, a selection of core subjects from various majors was carried out in several centers at the University of Extremadura. The questionnaire was distributed in April and May of the 2021–2022 academic year, in digital format, and it was available in the virtual classrooms of many professors who taught those subjects. Student responses were guaranteed total anonymity, so it is affirmed that the perceptions expressed are the fruit of their freedom of expression. Likewise, the students were not notified of the questionnaire's delivery nor were they given prior information about the subject to be discussed, which determines the non-existence of conditions, both of training in the subject and of the specific moment of delivery. The teachers for each subject contributed by encouraging students to complete it. The average duration spent by students answering the questionnaire was 10 min and the coding process was done through the design of common rules agreed upon by the research team. Once the questionnaires were completed correctly and anonymously, they were counted and classified according to the variables, obtaining a sample of 111 valid questionnaires (see Table 1).

In relation to the characteristics of the sample (Table 2), gender diversity has been considered, highlighting 54% male and 46% female. In relation to the age of the students, 85.6% are between the ages of 18 and 20, followed by 9.9% of students between the ages of 20 and 24. Regarding the major, 82.8% study Social Sciences, and 17.2% study Engineering and Mathematics. Specifically, in the social branch, students from the Business Administration and Management (BAM)-Law double-major, Economics, BAM and Labor Relations and Human Resources (LRHR) stand out, although the latter have little presence in the sample. In the branch of Engineering and Mathematics, the answers from Industrial, Mechanical and Electronic Engineering abound. No answers have been found from students studying other majors at the UEx (Education and Teacher Training, Philosophy and Letters, Medicine, Sciences, Veterinary, Nursing, Communication and Information Sciences, etc.). In relation to the highest year enrolled, 92.8% are first-year students and 3.6% are second-year. When asked if they have a scholarship to finance their studies, 52.3% stated yes and 47.7% replied no. When asked if they have their own monthly income, 90% replied no, 5.5% state to earn up to €600 per month and 4.5% earn more than €1000 per month. Finally, in relation to the campus where they are studying, 60.4% are in Badajoz, 34.2% in Cáceres, 4.5% in Plasencia and 0.9% in Mérida.

Regarding the study sample, the possible presence of three biases has been explored: social desirability bias, non-response bias and common method bias.

Firstly, in order to mitigate the social desirability bias, it was confirmed that the survey was carried out anonymously, following the approach of Fisher (1993), not having at any time asked the students for their identification or any data of a personal nature. As already reported in the questionnaire, the data would be treated in an aggregate manner, not individually.

Second, to address nonresponse bias, initial and subsequent responses were compared, dividing them into two groups. The first of them comprised 80% of the initial responses and the other the rest. An ANOVA was carried out, finding no significant differences between these groups, thus ruling out the existence of a non-response bias.

Thirdly, in order to evaluate the presence of bias, Harman's single factor test (Podsakoff, MacKenzie, Lee, & Podsakoff, 2003; Malhotra, Kim, & Patil, 2006) was performed as a diagnostic tool through Common Method Variance analysis (CMV). A Confirmatory Factor Analysis (CFA) was carried out using the IBM SPSS program v.26, and the reliability of each of the model constructs was subsequently examined through SEM. Two aspects will be indicative of the presence of CMV. On the one hand, when performing the CFA, if the results are grouped into a single factor. On the other hand, when a single factor explains most of the variance of the variables in the model. The results of the CFA indicated the existence of 3 factors, with a Kaiser-Mayer-Olkin (KMO) measure of sampling adequacy of 0.825, and a Bartlett's test of sphericity significant (8564.246, $p < 0.001$). The total of these factors represented 87.7% of the total variance explained in the model. The main factor explained only 41.2% of the variance, suggesting that common method variance bias is negligible in this study, making it unlikely that nonresponse bias exists in the proposed research model.

Table 1
Technical sheet of the study.

Universe of measurement and scope	First-year undergraduate students at the University of Extremadura
Number of responses	111
Sampling procedure	Random sampling
Type of population	Finite sample
Information collection method	Structured questionnaire done online
Data collection date	April and May 2022, 2021–2022 academic year

Source: The authors

Table 2
Participant profile.

Characteristics of the sample	n	%
<i>Gender diversity</i>		
- Male	60	54
- Female	51	46
<i>Student age</i>		
• Under 18 years	1	0.9
• 18–20 years	95	85.6
• 20–24 years	11	9.9
• 24–30 years	0	0
• Over 30 years	4	3.6
<i>Major</i>		
• Social Sciences (SS)		
BAM-Law	42	37.8
Economy	28	25.2
BAM	20	18
LRHR	2	1.8
Total SS	92	82.8
• Engineering and Mathematics (E and M)		
Industrial Engineering	6	5.4
Mechanical Engineering	6	5.4
Electronic Engineering	4	3.6
Mathematics	3	2.8
Total E and M	19	17.2
<i>Highest year in which they are enrolled</i>		
• 1st year	103	92.8
• 2nd year	4	3.6
• 3rd year	2	1.8
• 4th year	2	1.8
<i>Scholarship to finance studies</i>		
• Yes	58	52.3
• No	53	47.7
<i>Provision of own monthly income</i>		
• You do not have your own income	100	90
• Up to €600	6	5.5
• Up to €1000	0	0
• More than €1000	5	4.5
<i>Campus where you study</i>		
• Badajoz	67	60.4
• Cáceres	38	34.2
• Merida	1	0.9
• Plasencia	5	4.5

Source: The authors

3.3. Data analysis and measurement scales

The research model was analyzed using partial least squares structural equation modeling (PLS-SEM); more exactly, we used SmartPLS 4 software (Ringle, Wende, & Becker, 2022). The model is formed by latent or unobserved variables, which are inferred from indicators (Chin, 1998b), and considered as second-generation multivariate models. In these models, we include relationships between multiple predictor variables and criteria. This technique is appropriate for exploratory research, does not require data normality, and small samples can be used (Hair Jr, Black, Babin, Anderson, & Tatham, 2010).

All the variables are composite lower-order (LOC) constructs Model A. The scales focused on all the constructs were based on National FE Plan (Bank of Spain & CNMV (2008)) as well as contributions from the Bank of Spain and the National Stock Market Commission. The defined indicators had to be evaluated in all cases with a Likert scale from 1 to 5 (1: totally disagree to 5: totally agree). The total of 52 items was distributed among the constructs as follows: S was measured with 5 items; B was measured using 8 items; E was measured with 5 items; I was measured with 5 items; IN was measured with 4 items; Q was measured with 5 items; FE knowledge was measured with 11 items; and the RC was measured with 9 items.

These items make up the measurement instrument, developed by the authors based on the contributions of Bank of Spain & CNMV (2008), which has been previously validated and is provided in Annex I.

4. Results

4.1. Measurement model assessment

Measurement instrument reliability must be assessed, observing that each item's factor loadings (λ) assess the individual reliability of each item. The condition is that λ has to be greater than 0.707 ($\lambda > 0.707$) (Chin & Dibbern, 2010; Hair, Risher, Sarstedt, & Ringle,

2019). Loading values obtained a range from 0.701 to 0.901, exceeding the recommended minimum value (see Table 3). Also, since our constructs represent Composite LOC Mode A, the indicators that define them must be correlated, so we analyzed the correlations weights (Rigdon, 2016; Henseler, 2017).

We checked the internal consistency of each construct using Cronbach's alpha (CA) and the factor's composite reliability (CR) (Nunnally & Bernstein, 1994; Cronbach & Shavelson, 2004). CA's recommended minimum value is over 0.7. In this study, all CA values met the condition (Hair, Black, Babin, Anderson, & Tatham, 2006), confirming the scales' high reliability. Moreover, the CR analysis also produced acceptable values according to Nunnally (1978) and Vandenberg and Lance (2000), who recommend scores above 0.80 for advanced research (see Table 3). Therefore, the constructs' internal consistency was confirmed.

In order to assess the model's validity, we checked the construct's convergent and discriminant validity by calculating the average variance extracted (AVE) (Fornell & Larcker, 1981; Hair, Ringle, & Sarstedt, 2011), which must be greater than 0.5 (Hair, Black, Babin, Anderson, & Tatham, 2010). Our values met the recommended minimum, so the model constructs' convergent validity can be considered satisfactory.

Finally, the constructs' discriminant validity reveals the existence of differences between each construct and its items with respect to the other constructs and their items. To analyze discriminant validity, we used two criteria: Fornell and Larcker and heterotrait-monotrait ratio (HTMT). First, according to Fornell and Larcker's (1981) criterion, the square root of the AVE of each construct must be greater than the inter-construct correlations of the model. The values—shown in diagonal and bold in Table 4—for vertical and horizontal AVE are below the correlations between constructs (Henseler, Ringle, & Sarstedt, 2015; Roldán & Sánchez-Franco, 2012). The results confirm the model's discriminant validity according to Fornell and Larcker's criterion. Concurrently, discriminant validity was calculated based on the heterotrait-monotrait ratio (HTMT) (see Table 4). Henseler et al. (2015) and Roldán and Sánchez-Franco (2012) state that this ratio should be below 0.90. All the values included in Table 4 below meet the HTMT criterion. So, they up the recommended minimum and can be valid. Therefore, we confirm that all the constructs in this study meet the established discriminant validity criteria.

The model's nomogram could be constructed next, showing the path coefficients' signs, magnitude, significance and R^2 values for dependent variables (see Fig. 2).

Table 3
Evaluation of measurement model.

Composite LOC Mode A	Indicator	Loadings (λ)	Weights	Cronbach's alpha	CR	AVE
Savings (S)	S1	0.730	0.506	0.726	0.798	0.570
	S2	0.701	0.360			
	S3	0.828	0.457			
Budgets (B)	B2	0.794	0.347	0.800	0.870	0.626
	B4	0.816	0.309			
	B5	0.813	0.306			
	B7	0.740	0.303			
Expenses (E)	E1	0.821	0.351	0.782	0.859	0.605
	E2	0.733	0.291			
	E4	0.802	0.338			
	E5	0.751	0.302			
Incomes (I)	I3	0.901	0.692	0.705	0.831	0.711
	I4	0.782	0.482			
Investment (IN)	IN2	0.842	0.517	0.701	0.861	0.756
	IN3	0.896	0.630			
Quality of life (Q)	QL1	0.758	0.232	0.812	0.873	0.633
	QL2	0.849	0.342			
	QL3	0.805	0.262			
	QL4	0.767	0.420			
Financial education (FE)	FE1	0.835	0.133	0.911	0.928	0.618
	FE2	0.856	0.178			
	FE3	0.812	0.160			
	FE4	0.809	0.158			
	FE5	0.745	0.160			
	FE7	0.716	0.161			
	FE10	0.734	0.135			
	FE11	0.768	0.190			
Responsible consumption (RC)	RC1	0.719	0.213	0.894	0.917	0.613
	RC3	0.743	0.179			
	RC4	0.782	0.167			
	RC5	0.708	0.140			
	RC6	0.863	0.215			
	RC7	0.791	0.170			
	RC8	0.858	0.191			

Note(s): LOC: Lower-order construct; CR: Composite reliability; AVE: Average variance extracted.

Table 4

Constructs' discriminant validity (Fornell and Larcker's criterion below the main diagonal and HTMT ratio above the main diagonal).

Construct	B	E	FE	I	RC	IN	QL	S
B	0.791	0.690	0.574	0.387	0.725	0.612	0.279	0.593
E	0.547	0.778	0.722	0.523	0.716	0.782	0.524	0.486
FE	0.492	0.618	0.786	0.558	0.616	0.609	0.278	0.409
I	0.278	0.368	0.426	0.843	0.591	0.640	0.342	0.258
RC	0.634	0.610	0.569	0.447	0.783	0.716	0.341	0.566
IN	0.460	0.565	0.486	0.409	0.570	0.870	0.589	0.393
QL	0.254	0.417	0.258	0.232	0.306	0.453	0.796	0.335
S	0.401	0.351	0.333	0.168	0.427	0.262	0.245	0.755

Source: Authors.

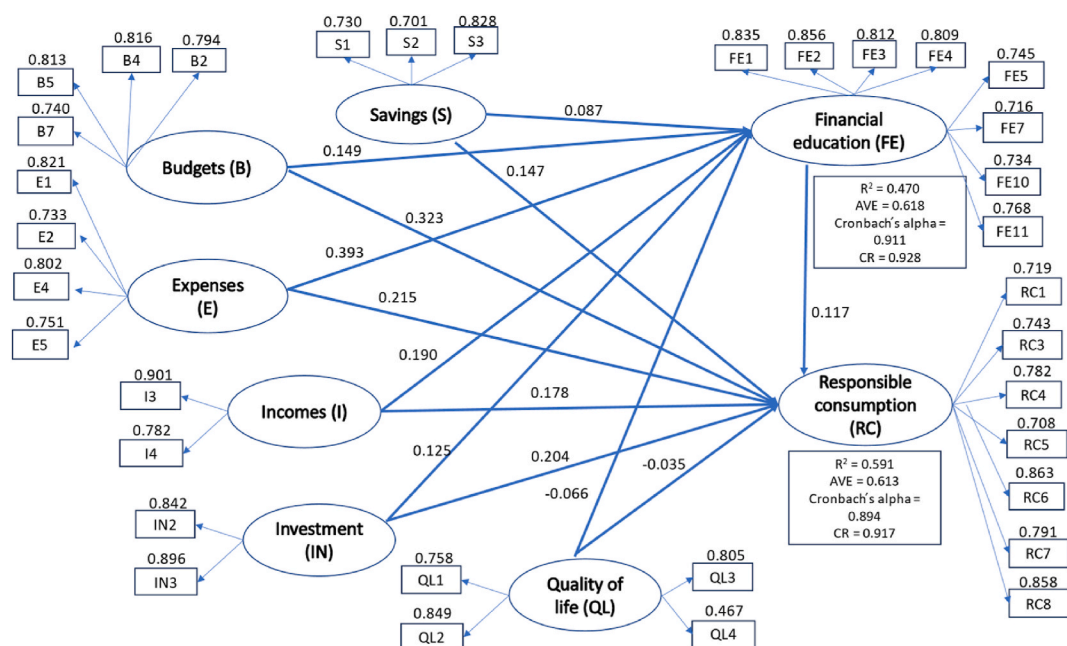
4.2. Structural model assessment

The structural model assesses the weight and magnitude of the relationships that define the research hypotheses (Wright, Campbell, Thatcher, & Robert, 2012). We analyzed the path coefficients' significance and if β coefficients are significant, the hypotheses are supported. For this, a *bootstrapping* procedure was applied with 5000 subsamples and a Student's *t-distribution*, based on a tail with $n - 1$ degrees of freedom, in which n is the number of subsamples (Chin, 1998a; Hair et al., 2011). In this study, the significant β values were 0.393*** (for H1c), 0.190* (for H1d), 0.323*** (for H2b) and 0.178* (for H2d), with nine values not meeting Chin's (1998b) recommended minimum. For H1c, H1d, H2b and H2d, the variables' significant positive effects were confirmed with p -value < 0.05 and $p < 0.001$. Moreover, this procedure provided the standard error and values of the Student's *t-statistic* for the parameters. Table 5 reveals that only four of the structural paths proposed in the model are significant, although with different levels of significance, and four hypotheses are supported by the results.

Finally, we analyzed the percentile confidence intervals (CI) and the bias corrected by applying the bootstrapping technique with 5000 subsamples. These values exceed a zero value for the case of relations that are satisfied, as recommended by Chin (1998a), thereby demonstrating the strength and the magnitude of the relationships of the constructs of the model (see Table 6).

4.3. Predictive capacity and model fit

A model's predictive capacity is determined according to each structural path's strength and analyzed by calculating the value of adjusted R^2 by the latent dependent variables (Chin, 2010). R^2 values must be equal to or greater than 0.1 for each path (Falk & Miller, 1992). In our case, R^2 varies between 0.47 for FE and 0.591 for RC, and the dependent constructs have R^2 values that exceed the minimum value (Table 7). On the other hand, we analyze the effect size through the value of f^2 . This measure assesses the relative

**Fig. 2.** Explanatory nomogram.

Source: Authors

Table 5
Hypotheses contrasted.

Hypothesis	Path coefficient (β)	P-value	Standard Error (SE)	t-value (bootstrap)	Decision (yes/no)
H1a: S $\rightarrow$ FE	0.087	0.296	0.083	1.045	Not accepted
H1b: B $\rightarrow$ FE	0.149	0.176	0.110	1.353	Not accepted
H1c: E $\rightarrow$ FE	0.393***	0.000	0.110	3.579	Accepted
H1d: I $\rightarrow$ FE	0.190*	0.047	0.096	1.985	Accepted
H1e: IN $\rightarrow$ FE	0.125	0.245	0.107	1.162	Not accepted
H1f: Q $\rightarrow$ FE	-0.066	0.501	0.097	0.673	Not accepted
H2a: S $\rightarrow$ RC	0.137	0.105	0.085	1.623	Not accepted
H2b: B $\rightarrow$ RC	0.323***	0.001	0.091	3.347	Accepted
H2c: E $\rightarrow$ RC	0.170	0.133	0.113	1.502	Not accepted
H2d: I $\rightarrow$ RC	0.178*	0.032	0.073	2.148	Accepted
H2e: IN $\rightarrow$ RC	0.190	0.141	0.129	1.472	Not accepted
H2f: Q $\rightarrow$ RC	-0.028	0.726	0.079	0.351	Not accepted
H3: FE $\rightarrow$ RC	0.117	0.253	0.102	1.143	Not accepted

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; based on a Student's $t_{(4999)}$ one-tailed distribution; $t_{(0.05, 4999)} = 1.645$; $t_{(0.01, 4999)} = 2.327$; $t_{(0.001, 4999)} = 3.092$.

Source: The authors

Table 6
Percentile CI and/or bias corrected CI.

Hypothesis	Percentile CI 5%.	Percentile CI 95%	Bias corrected CI 5%.	Bias corrected CI 95%
H1a: S $\rightarrow$ FE	-0.062	0.263	-0.091	0.233
H1b: B $\rightarrow$ FE	-0.052	0.384	-0.064	0.373
H1c: E $\rightarrow$ FE	0.169	0.599	0.187	0.617
H1d: I $\rightarrow$ FE	0.007	0.366	0.001	0.374
H1e: IN $\rightarrow$ FE	-0.089	0.334	-0.077	0.347
H1f: Q $\rightarrow$ FE	-0.264	0.117	-0.278	0.106
H2a: S $\rightarrow$ RC	-0.048	0.287	-0.047	0.287
H2b: B $\rightarrow$ RC	0.144	0.499	0.133	0.486
H2c: E $\rightarrow$ RC	-0.063	0.379	-0.063	0.378
H2d: I $\rightarrow$ RC	0.010	0.297	0.010	0.296
H2e: IN $\rightarrow$ RC	-0.093	0.412	-0.085	0.416
H2f: Q $\rightarrow$ RC	-0.191	0.118	-0.204	0.111
H3: FE $\rightarrow$ RC	-0.067	0.336	-0.085	0.314

Note: * $p < 0.05$; ** $p < 0.01$; *** $p < 0.001$; based on a Student's $t_{(4999)}$ one-tailed distribution; $t_{(0.05, 4999)} = 1.645$; $t_{(0.01, 4999)} = 2.327$; $t_{(0.001, 4999)} = 3.092$.

Source: The authors

importance of a predictor construct on an endogenous variable. The results show that the strongest relationships are E $\rightarrow$ FE = 0.154 (moderate effect) and B $\rightarrow$ RC = 0.138 (moderate effect).

Moreover, various indices available for PLS can be used to assess the model's goodness of fit (Henseler, 2017; Henseler, Hubona, & Ray, 2016). First, we can consider the standardized root mean square residual (SRMR) (Henseler et al., 2014). In this study, the SRMR was found to have a value of 0.087, so the model's overall fit is satisfactory. Second, the d_ULS and d_G indices can also be used. These fitness tests reach values of 4.553 and 1.844, respectively (i.e., lower than the 95% percentile), confirming that any discrepancies are not significant (see Table 7). Third, the correlation value of the mean square error (i.e., RMSttheta) was analyzed (Henseler, Hubona, & Ray, 2016). In the current research, this indicator achieved a value of 0.122. This indicator must be close to 0 and less than 0.12. The value obtained is, nonetheless, considered to be within the required margins (Henseler, Ringle, & Sarstedt, 2016) (see Table 7 above). The tests confirm that the global model is a good fit for the data.

5. Discussion and conclusions

This study covers a novel area of research as it focuses on analyzing the connection between compulsory education and university

Table 7
Model fit.

Constructs	R ² (explained variance)	SRMR	d_ULS	d_G	RMSttheta
FE	0.470				
RC	0.591	0.087	4.553	1.844	0.122

Note: R² = coefficient of determination.

Source: The authors

studies. Specifically, it studies the connection of financial knowledge acquired in compulsory education with the appreciation of the importance of FE and its applicability in daily life by establishing responsible, healthy, ethical, sustainable and supportive consumption habits. The research analyzes the perception that first-year university students have about the FE acquired in previous stages and the practical application in their personal lives. Specifically, we analyze whether these students have responsible consumption habits and whether these are related to the appreciation of the importance of FE and its attributes.

Interest in the study of the level of FE has grown considerably in recent years due to the absence of this knowledge (OECD, 2005; Lusardi & López, 2016; OECD/INFE, 2016; Lusardi et al., 2020) and the need for citizens to be prepared to make financial decisions (Santos, 2017; Moreira Costa et al., 2021). In view of this, concern about the world population's low level of FE has led numerous countries to implement national strategies that promote FE. This study arises from the concern expressed by the OECD (2005), which suggested the need to sufficiently anticipate knowledge of FE in order to acquire responsible consumption habits. Following this need, we consider it important to approach students in compulsory education. However, FE should be understood as a lifelong learning process (Villalba et al., 2017; Gallardo-Vázquez et al., 2023). Therefore, it is essential to take into account the necessary connection between educational stages.

Our interest in this level of teaching has been preceded by other studies focused on various countries and with different approaches. Studies on FE in the United States stand out (Bernheim, Garrett, & Maki, 2001, 1998; Braunstein & Welch, 2002), along with others focused on the Big Three (Lusardi and Mitchell, 2007, 2011a, 2011b, 2014) and clearly highlighting stakeholders with a low level of knowledge of FE (Lusardi et al., 2020). Likewise, the connection between the level of FE and aspects such as investment, savings, budget and quality of life has been taken into account (Van Rooij et al., 2011; de Bassa Scheresberg, 2013; Jappelli & Padula, 2013; Atkinson et al., 2015; Moreira Costa et al., 2021).

The students' perception of FE and responsible consumption rested on the formulation of three research questions. The first of them was aimed at observing whether (RQ1) *Do students identify the fundamental attributes that determine their level of knowledge in FE?* Of the six hypotheses defined to support it, only two were supported. We cannot therefore ensure that students completely and correctly identify the fundamental attributes that are determining their level of knowledge of FE. The second question was aimed at observing whether (RQ2) *Do students identify the attributes of responsible consumption?* Similarly, six hypotheses were defined and only two were supported. This tells us that it cannot be ensured that students correctly identify the attributes that determine responsible consumption. The third question was aimed at observing whether (RQ3) *Do students relate the level of FE with the responsible consumption they should do?* For this, a hypothesis was defined that could not be confirmed either.

A joint analysis tells us that thirteen hypotheses were defined linking the independent constructs (savings, budget, expenses, income, investment and quality of life) with the two dependent constructs (financial education and responsible consumption). Not all were supported. Specifically, only H1c: $E \rightarrow FE$, H1d: $I \rightarrow FE$, H2b: $B \rightarrow RC$ and H2d: $I \rightarrow RC$ were accepted, confirming that better knowledge and attitude regarding expenses and income are associated with a greater perception of the importance of FE. It is also confirmed that better knowledge and attitude regarding budget and income are associated with a greater ability to consume responsibly.

In relation to the unsupported hypotheses, they determine the need for further work on such relationships. On the one hand, in order to achieve the fulfillment of H1a: $S \rightarrow FE$, H1b: $B \rightarrow FE$, H1e: $I \rightarrow FE$ and H1f: $Q \rightarrow FE$, it would be necessary to promote knowledge of the concepts of savings, budget, investment and quality of life, which would allow us to confirm that a better knowledge and attitude with respect to them would be associated with a greater perception of the importance of FE. On the other hand, in order to achieve the fulfillment of H2a: $S \rightarrow RC$, H2c: $E \rightarrow RC$, H2e: $IN \rightarrow RC$ and H2f: $Q \rightarrow RC$, it would be necessary to promote knowledge of the concepts of saving, spending, investment and quality of life, which would allow us to confirm that better knowledge and attitude regarding them would be associated with a greater ability to consume responsibly. Finally, in order to achieve the fulfillment of H3: $FE \rightarrow RC$, it would be necessary to acquire a greater perception of the importance of FE, which could be associated with a greater capacity to consume responsibly.

These results have relevant practical implications for the university community and for the regional authorities responsible for secondary and primary education. This study will make it possible to propose lines of action of a transversal nature for university and pre-university studies. Those responsible for education, at the regional, national, and international levels, should devise internal actions that will provide them with the adequate promotion of FE and the practice of RC. At the same time, they must work for these practices to be adopted by society and to be distributed in such a way that guarantees the incorporation of knowledge and the adoption of the practices, as well as ensuring they reach as much of the population as possible. In short, it is clear that FE can be a useful tool for creating responsible and sustainable behaviors among young people, resulting in future economic and social prosperity.

Although this study contributes to the expansion of the literature on FE and RC, is not free of limitations. The responses were obtained as a subjective expression of the student collective, which could lead to some bias in the research. We also mentioned the research's regional focus, which may determine participation biases motivated by cultural and other differences derived from the regional assumption of competencies in education. Given these limitations, future extensions of the study are proposed, expanding the sample with students from other regions and countries, designing a questionnaire that is more universal than the current one, and including other variables and artificial intelligence.

CRediT authorship contribution statement

Dolores Gallardo-Vázquez: Writing – original draft, Visualization, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **José Luis Miralles-Quirós:** Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Conceptualization. **María Mar Miralles-**

Quirós: Writing – review & editing, Writing – original draft, Validation, Software, Project administration, Methodology, Investigation, Formal analysis, Conceptualization.

Conflict of interest

The authors declare no potential conflicts of interest with respect to the research, authorship, and publication of this paper.

Funding

Open Access funding enabled and organized by Universidad de

Declaration of competing interest

None.

Annex I

1 – Totally disagree and 5 – Totally agree

PERCEPTION OF MY KNOWLEDGE OF FINANCIAL EDUCATION (FE)	1	2	3	4	5
FE will allow me to manage my money					
FE will allow me to plan for the future					
FE will help me make personal financial decisions in my life					
FE will help me choose and use financial products and services					
FE will allow me to access information related to economic and financial matters					
FE will help me to be honest, using my own and other people's money correctly					
FE will help me to be responsible, paying for what I want to buy					
FE will help me to be reliable, returning borrowed money					
FE will help me to be supportive, contributing when needs arise					
FE will help me to be able to foster an entrepreneurial spirit					
FE will allow me to acquire knowledge to function in the world of finance					
WE CONSIDER IT NECESSARY TO STRENGTHEN KNOWLEDGE, SKILLS AND ABILITIES OF AN ECONOMIC NATURE	1	2	3	4	5
SAVING					
I consider savings to be the part of a period's income that I have not spent					
I decide to save when I have to face unexpected expenses					
I decide to save when I want to buy a good that I like or need					
I decide to save when I want to invest and obtain a return					
I decide to save to help disadvantaged groups					
BUDGET					
I consider the budget as a list of expenses and income expected for a period of time					
I must know how much our income is and how much we are going to spend					
I must know what I am going to spend each euro on					
I must have a reserve of money for unforeseen expenses					
I must control the accounts					
The budget helps to reduce or eliminate possible debts					
The budget allows for making future forecasts					
The budget will allow me to get the most out of my money					
EXPENSE					
Organizing my expenses is essential					
I respect the expenses identified in the budget					
In the event of an unexpected expense, I could know where I get the funds from					
I consider it important to create a record of my expenses					
Rational use of consumption will allow me to reduce the expenses necessary to live					
INCOME					
Are you worried about having an income at this stage of your life?					
Would you like to have an income to cover your regular expenses?					
Would you like to have an income to travel?					
Would you like to have an income to rent or buy a home?					
My income allows me the luxury of eating out, going to the movies, going out with friends					
INVESTMENT					
I consider it to be risking my savings in the hope of earning more in return					
I know the concept of profitability					
It is important to know financial goals and know how to achieve them					
I am able to invest in financial assets					
QUALITY OF LIFE					
The training I have in financial education will allow me to have a good quality of life					
The training I have allows me to use my income well					

(continued on next page)

(continued)

The training I have allows me to make good use of my savings in the expenses I have
 The budget I make for myself each month allows me to have an acceptable quality of life
 Living day to day does not give me a good quality of life
SMART, RESPONSIBLE, HEALTHY, ETHICAL, SUSTAINABLE AND SOLIDARY CONSUMPTION
 I know that consumption consists of using products and services to satisfy our needs and desires
 I consider that the intelligent consumer is one who knows what he needs
 I consider that the intelligent consumer is one who chooses the appropriate products and services for quality and price
 I consider that the intelligent consumer is one who minimizes the impact of his consumption on natural resources and the environment
 I agree to consume with environmental conservation in mind
 I agree to consume products that improve health
 I agree to consume with the most fair, supportive and ecological social implications in mind
 I agree to consume efficiently, which satisfies the needs of the present and does not harm future needs
 I agree with consuming products from poor countries and groups with social needs

Data availability

Data will be made available on request.

References

- Allgood, S., & Walstad, W. (2011). The effects of perceived and actual financial knowledge on credit card behavior. *Networks Financial Institute Working*, 15, 1–26. <https://doi.org/10.2139/ssrn.1896365>. Paper No. 15.
- Atkinson, A., & Messy, F. (2013). Promoting financial inclusion through financial education: OECD/INFE evidence, policies and practice. *OECD working papers on finance, insurance and private pensions*, No. 34. OECD Publishing. <https://doi.org/10.1787/5k3xz6m88smp-en>
- Atkinson, A., Messy, F. A., Rabinovich, L., & Yoong, J. (2015). Financial education for long-term savings and investments: Review of research and literature. *OECD working papers on finance, insurance and private pensions* No. 39.
- Bank of Spain & CNMV. (2008). Plan de Educación Financiera 2008-2012. <https://www.cnmv.es/DocPortal/Publicaciones/PlanEducacion/PlanEducacion.pdf>.
- Barbić, D., Lučić, A., & Chen, J. M. (2019). Measuring responsible financial consumption behaviour. *International Journal of Consumer Studies*, 43, 102–112. <https://doi.org/10.1111/ijcs.12489>
- Bernheim, B. D., Garrett, D. M., & Maki, D. M. (2001). Education and saving: The long-term effects of high school financial curriculum mandates. *Journal of Public Economics*, 80(3), 435–465. [https://doi.org/10.1016/S0047-2727\(00\)00120-1](https://doi.org/10.1016/S0047-2727(00)00120-1)
- Braunstein, S., & Welch, C. (2002). Financial literacy: An overview of practice, research, and policy. *Federal Reserve Bulletin*, 446–457. <https://www.federalreserve.gov/pubs/bulletin/2002/1102lead.pdf>.
- CFSI. (2016). *Eight ways to measure financial health*. Center for Financial Services Innovation working paper.
- Chin, W. W. (1998a). The partial least squares approach to structural equation modeling. In G. A. Marcoulides (Ed.), *Modern methods for business research* (pp. 295–336). Hillsdale, NJ: Lawrence Erlbaum.
- Chin, W. W. (2010). How to write up and report PLS analyses. In V. Esposito, W. W. Chin, J. Henseler, & H. Wang (Eds.), *Handbook of partial least squares: Concepts, methods and applications in marketing and related fields* (pp. 655–690). Berlin: Springer.
- Chin, W. W., & Dibbern, J. (2010). An introduction to a permutation based procedure for multi-group PLS analysis: Results of tests of differences on simulated data and a cross cultural analysis of the sourcing of information system services between Germany and the USA. In V. Esposito, W. W. Chin, J. Henseler, & H. Wang (Eds.), *Handbook of partial least squares: Concepts, methods and applications in marketing and related fields* (pp. 171–193). Berlin: Springer.
- Cronbach, L. J., & Shavelson, R. J. (2004). My current thoughts on coefficient alpha and successor procedures. *Educational and Psychological Measurement*, 64(3), 391–418. <https://doi.org/10.1177/0013164404266386>
- Cull, R., Ehrbeck, T., & Holle, N. (2014). Financial inclusion and development: Recent impact evidence. CGAP focus note 92 World Bank Group <http://documents.worldbank.org/curated/en/269601468153288448/Financial-inclusion-and-development-recent-impact-evidence>.
- de Bassa Scheresberg, C. (2013). Financial literacy and financial behavior among young adults: Evidence and implications. *Numeracy*, 6(2), 5. <https://doi.org/10.5038/1936-4660.6.2.5>
- Domínguez, J. M. (2017). Los programas de educación financiera: aspectos básicos y referencia al caso español. *Revista electrónica sobre la enseñanza de la Economía Pública*, 20, 19–60.
- EESC. (2016). Educación financiera para todos. *Estrategias y buenas prácticas de educación financiera en la Unión Europea*. <https://www.eesc.europa.eu/sites/default/files/resources/docs/qe-30-12-894-es.pdf>.
- European Commission. (2007). Communication from the commission. Financial Education. Brussels, 18.12.2007 COM (2007) 808 final <https://eur-lex.europa.eu/LexUriServ/LexUriServ.do?uri=COM:2007:0808:FIN:EN:PDF>.
- Falk, R. F., & Miller, N. B. (1992). *A primer for soft modeling*. Akron, OH: University of Akron Press.
- Fisher, R. J. (1993). Social desirability bias and the validity of indirect questioning. *Journal of Consumer Research*, 20(2), 303–315. <https://doi.org/10.1086/209351>
- Fornell, C., & Larcker, D. F. (1981). Evaluating structural equation models with unobservable variables and measurement error. *Journal of Marketing Research*, 18(1), 39–50. <https://www.jstor.org/stable/3151312>.
- Gale, W. G., & Levine, R. (2011). Financial literacy: What works? How could it be more effective?. https://papers.ssrn.com/sol3/papers.cfm?abstract_id=2316933.
- Gallardo-Vázquez, D., Miralles-Quiros, M. M., & Miralles-Quiros, J. L. (2023). The perception about financial education among students at the university of Extremadura. In *The impact of HELs on regional development: Facts and practices of collaborative work with SMEs* (pp. 293–317).
- Gallego-Losada, R., Montero-Navarro, A., Rodríguez-Sánchez, J. L., & González-Torres, T. (2022). Retirement planning and financial literacy, at the crossroads. A bibliometric analysis. *Finance Research Letters*, 44(January 2022), Article 102109. <https://doi.org/10.1016/j.frl.2021.102109>
- Garg, N., & Singh, S. (2018). Financial literacy among youth. *International Journal of Social Economics*, 45(1), 173–186. <https://doi.org/10.1108/IJSE-11-2016-0303>
- Gascó, R., & Estruch, J. (2022). *La Educación Financiera del alumnado de educación secundaria*. Eufinet working paper.
- Goyal, K., & Kumar, S. (2021). Financial literacy: A systematic review and bibliometric analysis. *International Journal of Consumer Studies*, 45(1), 80–105. <https://doi.org/10.1111/ijcs.12605>
- Hair, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2006). *Multivariate data analysis* (Vol. 6). Upper Saddle River, NY: Pearson Prentice Hall.
- Hair Jr, J. F., Black, W. C., Babin, B. J., Anderson, R. E., & Tatham, R. L. (2010). Sem: An introduction. In J. F. Hair, W. C. Black, B. J. Babin, & R. E. Anderson (Eds.), *Multivariate data analysis: A global perspective* (pp. 629–686). Upper Saddle River, NY: Pearson Prentice Hall.
- Hair, J. F., Ringle, C. M., & Sarstedt, M. (2011). PLS-SEM: Indeed a silver bullet. *Journal of Marketing Theory and Practice*, 19(2), 139–152. <https://doi.org/10.2753/MT1069-6679190202>

- Hair, J. F., Risher, J. J., Sarstedt, M., & Ringle, C. M. (2019). When to use and how to report the results of PLS-SEM. *European Business Review*, 31, 2–24. <https://doi.org/10.1108/EBR-11-2018-0203>
- Henseler, J. (2017). Bridging design and behavioral research with variance-based structural equation modeling. *Journal of Advertising*, 46, 178–192.
- Henseler, J., Dijkstra, T. K., Sarstedt, M., Ringle, C. M., Diamantopoulos, A., Straub, D. W., ... Calantone, R. J. (2014). Common beliefs and reality about PLS comments on Rönkkö and Evermann (2013). *Organizational Research Methods*, 17(2), 182–209. <https://doi.org/10.1177/1094428114526928>
- Henseler, J., Hubona, G., & Ray, P. A. (2016). Using PLS path modeling in new technology research: Updated guidelines. *Industrial Management and Data Systems*, 116, 2–20. <https://doi.org/10.1108/IMDS-09-2015-0382>
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2015). A new criterion for assessing discriminant validity in variance-based structural equation modeling. *Journal of the Academy of Marketing Science*, 43(1), 115–135. <https://doi.org/10.1007/s11747-014-0403-8>
- Henseler, J., Ringle, C. M., & Sarstedt, M. (2016). Testing measurement invariance of composites using partial least squares. *International Marketing Review*, 33, 405–431.
- Jappelli, T., & Padula, M. (2013). Investment in financial literacy and saving decisions. *Journal of Banking & Finance*, 37(8), 2779–2792. <https://doi.org/10.1016/j.jbankfin.2013.03.019>
- Johan, I., Rowlingson, K., & Appleyard, L. (2021). The effect of personal finance education on the financial knowledge, attitudes and behaviour of university students in Indonesia. *Journal of Family and Economic Issues*, 42(2), 351–367. <https://doi.org/10.1007/s10834-020-09721-9>
- LaBorde, P. M., Mottner, S., & Whalley, P. (2013). Personal financial literacy: Perceptions of knowledge, actual knowledge and behavior of college students. *Journal of Financial Education*, 39(3/4), 1–30. <https://www.jstor.org/stable/23608645>.
- Lusardi, A. (2014). *Financial literacy: An essential tool for informed consumer choice?* NBER working paper.
- Lusardi, A., & López, A. (2016). *Financial literacy among high school students in the United States: Evidence from the 2012 programme for international student assessment (PISA)*. European Investment Bank Institute Working Paper.
- Lusardi, A., Michaud, P. C., & Mitchell, O. S. (2020). Assessing the impact of financial education programs: A quantitative model. *Economics of Education Review*, 78, 1–13. <https://doi.org/10.1016/j.econedurev.2019.05.006>
- Lusardi, A., & Mitchell, O. S. (2007). Financial literacy and retirement preparedness: Evidence and implications for financial education. *Business Economics*, 42(1), 35–44. <https://doi.org/10.2145/20070104>
- Lusardi, A., & Mitchell, O. S. (2011a). Financial literacy and planning: Implications for retirement well-being. In A. Lusardi, & O. S. Mitchell (Eds.), *Financial literacy: Implications for retirement security and the financial marketplace*. Oxford University Press.
- Lusardi, A., & Mitchell, O. S. (2011b). Financial literacy around the world: An overview. *Journal of Pension Economics and Finance*, 10(4), 497–508. <https://doi.org/10.1017/S1474747211000448>
- Lusardi, A., & Mitchell, O. S. (2014). The economic importance of financial literacy: Theory and Evidence. *Journal of Economic Literature*, 52(1), 5–44. <https://doi.org/10.1257/jel.52.1.5>
- Lusardi, A., & Tufano, P. (2015). Debt literacy, financial experiences, and overindebtedness. *Journal of Pension Economics and Finance*, 14(4), 332–368. <https://doi.org/10.1017/S1474747215000232>
- Malhotra, N. K., Kim, S., & Patil, A. (2006). Common method variance in IS research: A comparison of alternative approaches and a reanalysis of past research. *Management Science*, 52, 1865–1883. <https://doi.org/10.1287/mnsc.1060.0597>
- Mandell, L., & Schmid, K. (2009). The impact of financial literacy education on subsequent financial behavior. *Journal of Financial Counseling and Planning*, 20(1), 15–24.
- Molina-García, A., Cisneros-Ruiz, A. J., López-Subieres, M. D., & Diéguez-Soto, J. (2023b). How does financial literacy influence undergraduates' risk-taking propensity? *International Journal of Management in Education*, 21(3), Article 100840. <https://doi.org/10.1016/j.ijme.2023.100840>
- Molina-García, A., Diéguez-Soto, J., Galache-Laza, M. T., & Campos-Valenzuela, M. (2023a). Financial literacy in SMEs: A bibliometric analysis and a systematic literature review of an emerging research field. *Review of Managerial Science*, 17(3), 787–826. <https://doi.org/10.1007/s11846-022-00556-2>
- Moreira Costa, V., De Sá Teixeira, N. A., Cordeiro Santos, A., & Santos, E. (2021). When more is less in financial decision-making: Financial literacy magnifies framing effects. *Psychological Research*, 85, 2036–2046. <https://doi.org/10.1007/s00426-020-01372-7>
- Moreno-Herrero, D., Salas-Velasco, M., & Sánchez-Campillo, J. (2018a). Factors that influence the level of financial literacy among young people: The role of parental engagement and students' experiences with money matters. *Children and Youth Services Review*, 95, 334–351. <https://doi.org/10.1016/j.childyouth.2018.10.042>
- Moreno-Herrero, D., Salas-Velasco, M., & Sánchez-Campillo, J. (2018b). The knowledge and skills that are essential to make financial decisions: First results from PISA 2012. *FinanzArchiv: Public Finance Analysis*, 74(3), 293–339. <https://doi.org/10.1628/fa-2018-0009>
- Muñoz-Céspedes, E., Ibar-Alonso, R., & de Lorenzo Ros, S. (2021). Financial literacy and sustainable consumer behavior. *Sustainability*, 13, 9145. <https://doi.org/10.3390/su13169145>
- Nunnally, J. (1978). *Psychometric methods*. New York: McGraw-Hill.
- Nunnally, J. C., & Bernstein, I. H. (1994). *Psychometric theory* (3 ed.). New York: McGraw-Hill.
- OECD. (2005). *Improving financial literacy: Analysis of issues and policies*. OECD Publishing. Paris <https://www.oecd.org/finance/financial-education/improvingfinancialliteracyanalysisissuesandpolicies.htm>.
- OECD/INFE. (2016). *International survey of adult financial literacy competencies*. Paris: OECD Publishing. <https://www.oecd.org/daf/fin/financial-education/OECD-INFE-International-Survey-of-Adult-Financial-Literacy-Competencies.pdf>.
- Ortega, M., Pino, C., Merino, M., & Ladrado, M. (2009). Educación Financiera en Enseñanza Secundaria Obligatoria Nivel I Guía para el profesorado. <https://www.oecd.org/finance/financial-education/50236319.pdf>.
- Podsakoff, P. M., MacKenzie, S. B., Lee, J.-Y., & Podsakoff, N. P. (2003). Common method biases in behavioral research: A critical review of the literature and recommended remedies. *Journal of Applied Psychology*, 88(5), 879–903. <https://doi.org/10.1037/0021-9010.88.5.879>
- Preuss, L., Elms, H., Kurdyukov, R., Golob, U., Zaharia, R. M., Jalsenjak, B., et al. (2023). Taking stock of responsible management education in Central and Eastern Europe. *International Journal of Management in Education*, 21(3). <https://doi.org/10.1016/j.ijme.2023.100889>
- Raccanello, K., & Herrera Guzmán, E. (2014). Educación e inclusión financiera. *Revista Latinoamericana de Estudios Educativos*, 44(2), 119–141.
- Razen, M., Huber, J., Hueber, L., Kirchler, M., & Stefan, M. (2021). Financial literacy, economic preferences, and adolescents' field behavior. *Finance Research Letters*, 40(May 2021). <https://doi.org/10.1016/j.frl.2020.101728>. Article 101728.
- Ringle, C. M., Wende, S., & Becker, J. M. (2022). SmartPLS 4. Ooststeinbek: SmartPLS. Retrieved from <https://www.smartpls.com>.
- Roldán, J. L., & Sánchez-Franco, M. J. (2012). Variance-based structural equation modeling: Guidelines for using partial least squares in information systems research. In M. Mora, O. Gelman, A. Steenkamp, & M. S. Raisinghani (Eds.), *Research methodologies, innovations and philosophies in software systems engineering and information systems* (pp. 193–221). USA: Hershey, PA.
- Samper, M. J., Pinto, C., Arenas, J., & Roca, M. (2010). *Mis finanzas personales*. Aguilar.
- Santos, A. C. (2017). Cultivating the self-reliant and responsible individual: The material culture of financial literacy. *New Political Economy*, 22(4), 410–422. <https://doi.org/10.1080/13563467.2017.1259302>
- Song, S. Y., & Kim, Y. K. (2018). Theory of virtue ethics: Do consumers' good traits predict their socially responsible consumption? *Journal of Business Ethics*, 152, 1159–1175. <https://doi.org/10.1007/s10551-016-3331-3>
- Standard & Poor's. (2015). Financial literacy around the world: Insights from the standard & poor's ratings services. City: Global financial literacy survey. <https://gflec.org/wp-content/uploads/2015/11/3313-Finlit-Report-FINAL-5.11.16.pdf?x53159>.
- Sutter, M., Weyland, M., Untertrifaller, A., & Froitzheim, M. (2020). Financial literacy, risk and time preferences - results from a randomized educational intervention. *SSRN Electronic Journal*. <https://doi.org/10.2139/ssrn.3676095>
- Van Rooij, M., Lusardi, A., & Alessie, R. (2011). Financial literacy and stock market participation. *Journal of Finance and Economics*, 10(2), 449–472. <https://doi.org/10.1016/j.jfineco.2011.03.006>

- Vandenberg, R. J., & Lance, C. E. (2000). A review and synthesis of the measurement invariance literature: Suggestions, practices, and recommendations for organizational research. *Organizational Research Methods*, 3(1), 4–70. <https://doi.org/10.1177/109442810031002>
- Villalba, F., López-Lezama, J. M., & Muñoz-Galeano, N. (2017). El Papel de la Educación Financiera en la Formación de Profesionales de la Ingeniería. *Formación Universitaria*, 10(2), 13–22. <https://doi.org/10.4067/S0718-50062017000200003>
- Wihlenda, M., Brahm, T., & Habisch, A. (2023). Responsible management education: Social entrepreneurial competences of civically-engaged students. *International Journal of Management in Education*, 21, Article 100756. <https://doi.org/10.1016/j.ijme.2022.100756>
- Wright, R. T., Campbell, D. E., Thatcher, J. B., & Robert, N. (2012). Operationalizing multidimensional constructs in structural equation modeling: Recommendations for IS research. *Communications of the Association for Information Systems*, 30, 367–412.
- Xiao, J. J., Chen, C., & Sun, L. (2015). Age differences in consumer financial capability. *International Journal of Consumer Studies*, 39(4), 387–395. <https://doi.org/10.1111/ijcs.12205>
- Xiao, J. J., & Dew, J. (2011). The financial management behavior scale: Development and validation. *Journal of Financial Counseling and Planning*, 22(1), 43–59.